3D MODEL

The **3D model** plugin can be used to display models from a **.gltf** or **.glb** file in the Layout View, along with playing their animations.

Check the 3D model object manual entry for information about how to import model files.

Refer to the 3D shape manual entry for information about working with 3D in Construct 3 which is also relevant to this plugin.

Scripting

When using JavaScript or TypeScript coding, the features of this object can be accessed via the I3DModelInstance script interface.

Collisions

Similarly to 3D Shape collisions for 3D models still happen in 2D.

3D model properties

3D Model
The 3D model object the instance will use.

Initially visible
If the model should be visible or not when the layout starts.

X offset
The offset in the X axis in relation to the corresponding instance.

Y offset
The offset in the Y axis in relation to the corresponding instance.

Z offset
The offset in the Z axis in relation to the corresponding instance.

X rotation
The rotation in the X axis.

Y rotation
The rotation in the Y axis.

Z rotation
The rotation in the Z axis. This is added to the angle of the corresponding instance.

X scale
The scale in the X axis. This is compounded with the scale of the corresponding instance.

Y scale
The scale in the Y axis. This is compounded with the scale of the corresponding instance.

Z scale
The scale in the Z axis. This is compounded with the scale of the corresponding instance.

Mesh
Choose which mesh to show. A single one or all of them.

Animation
The initial animation.

Mesh render mode
The mesh render mode. Can be *"hierarchy"* to draw meshes using the hierarchy defined in the model file, *"isolate"* will draw the enabled models as if they where the only ones in the model file.

Initially playing
If the model should start playing an animation or not when the layout starts.

Initial animation progress
The initial animation progress.

Enable collisions
If the instance should collide or not.

Bounding box
An editor only property to display the bounding box of the 3D model.

Bounding box color
An editor only property to change the color of the bounding box of the 3D model.

3D model conditions

For conditions in common to other objects, see Common conditions.

On model loaded
Triggered when a new 3D model object is loaded.

On model load error
Triggered when there is an error loading a 3D model object.

For each mesh
Loop through all the meshes in the 3D model.

For each animation
Loop through all the animations in the 3D model.

Is playing
Checks if there is an animation playing.

Render mode
The render mode indicates how the meshes are displayed. If *"hierarchy"* is used, all the displayed meshes are drawn following the authored hierarchy in the model file. If *"isolate"* is used, the displayed meshes are drawn as if they where the only ones in the model file.

Is mesh enabled
Checks if the provided mesh name is being drawn or not.

Are all meshes enabled
Checks if all the meshes in the model are being drawn.

Mesh exists
Checks if a mesh exists in the model.

Enable collisions
Checks if collisions are enabled.

3D model actions

For actions common to other objects, see Common actions.

Set 3D model
Start loading a new 3D model object. The "**Mesh**" parameter can optionally be used to draw a single mesh, if the parameter is left empty all the meshes will be drawn. Use the "**Animation**" parameter to choose the starting animation, if left empty the first animation is used. Choose if the animation should start playing using the "**Playing**" parameter and the "**Progress**" parameter to choose where the animation should start from, using a value in the range **[0 - 1]**.

Set 3D model (by name)
This is the same as "**Set 3D model**" but the 3D model object is specified using a string.

Set mesh enabled
Choose if a mesh is drawn or hidden.

Set all meshes enabled
Choose if all meshes are drawn or hidden.

Set animation
Change the current animation. Choose if the animation should play using the "**Playing**" parameter and use the "**Progress**" parameter to choose where the animation should start from, using a value in the range **[0 - 1]**.

Set progress
Set the progress of the current animation using a value in the range **[0 - 1]**

Set transform
Change the transformation values of the model. "**Transform X**", "**Transform Y**", "**Transform Z**" correspond to each axis. Use the "**Operation**" parameter to choose which operation to do on the values, either *Set*, *Add*, *Subtract*, *Multiply* or *Divide*. Use the "**Type**" parameter to choose which transformation to affect, either *Offset*, *Rotation* or *Scale*.

Play
Start playing the specified animation and use the "**Progress**" parameter to choose where the animation should start from, using a value in the range **[0 - 1]**

Stop
Stop the current animation. The progress is reset to 0.

Pause
Pause the current animation. The progress stays as is.

Resume
Resume a paused animation.

Set mesh render mode
Change the render mode of the model. Either *"hierarchy"* or *"isolate"*.

Set collisions enabled
Change the collision state.

3D model expressions

For expressions common to other objects, see common expressions.

Error3dModelName
The name of the last 3D model object that failed to load.

Name
The name of the 3D model object.

Meshes
A comma separated string with all the currently enabled mesh names.

Animation
The current animation.

AnimationDuration(Animation)
The total duration of the specified animation.

Progress
The progress of the current animation as a value in the range **[0 - 1]**.

MeshCount
The amount of meshes in the current model.

MeshAt(Index)
The mesh at the specified index.

AnimationCount
The total amount of animations in the model.

AnimationAt(Index)
The animation at the specified index.

CurMesh
The current mesh in the "**For each mesh**" looping condition.

CurAnimation
The current animation in the "**For each animation**" looping condition.

OffsetX
The offset in the X axis in relation to the corresponding instance.

OffsetY
The offset in the Y axis in relation to the corresponding instance.

OffsetZ
The offset in the Z axis in relation to the corresponding instance.

RotationX
The rotation in the X axis.

RotationY
The rotation in the Y axis.

RotationZ
The rotation in the Z axis. This value is added to the angle of the corresponding instance.

ScaleX
The scale in the X axis. This is compounded with the scale of the corresponding instance.

ScaleY
The scale in the Y axis. This is compounded with the scale of the corresponding instance.

ScaleZ
The scale in the Z axis. This is compounded with the scale of the corresponding instance.

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This manual entry was last updated on 12 Mar, 2026 at 18:55